- **Cytokines**

Cytokines are proteins made by cells that affect the behavior of other cells. When released from mononuclear phagocytes, they are called monokines. When released from T lymphocytes they are called lymphokines. When released from leukocytes they are called interleukins. Cytokines are required for regulation of both the nonspecific and specific immune responses. Interferons (IFNS) are a group of cytokines produced by virus infected cells. Several classes of interferons are recognized. IFN γ is synthesized by virus infected leukocytes, antigen stimulated T cells and natural killer cells. IFN α / β is derived from virus infected fibroblasts. Interferons prevent viral replication and assembly, thereby limiting viral infection.

Another group of noteworthy cytokines are endogenous pyrogens which elicit fever in the host. Examples of endogenous pyrogens include interleukin – 1, Interleukin – 6 and tissue necrosis factor. All are produced by host macrophages in response to pathogens.

- **Complement system**

The complement system is a part of the immune system, consists of a series of proteins that interact with one another in a highly regulated manner, in order to eliminate pathogens. Complements are soluble proteins and glycoproteins mostly produced by hepatocytes. More than 20 types of complements are present in serum found circulating normally in human body in inactive forms (called as zymogens or proenzymes). Complement activation is triggered by an antibody when it is bound to the antigen. It can also be triggered by some components of innate immunity. Thus the complement system works in both innate and acquired immunity.

Functions of complements

Some major functions of complements are:

- Opsonization and phagocytosis
- Cell lysis
- Chemotaxis
- Activation of mast cells and basophils and enhancement of inflammation
- Production of antibodies
- Immune clearance and inflammation by attracting macrophages and neutrophils.

C. Phagocytosis

i. Phagocytosis is the ingestion by phagocytic cells of invading foreign particles such as bacteria. After ingestion, the foreign particle is entrapped in a phagocytic vacuole (phagosome), which fuses with lysosomes forming the phagolysosome. The lysosomes release their powerful lytic enzymes which digest the particle. (Figure 13.19). Phagocytosis is conducted by blood monocytes, neutrophils and tissue macrophages. Phagocytosis may be enhanced by a variety of factors collectively referred to as opsonins which consist of antibodies and various serum components of complement.

ii. Phagocytic cells use two basic mechanisms for the recognition of microorganisms. Opsonin dependent and opsonin independent

iii. Phagocytes use pathogen recognition receptors to detect pathogen associated molecular patterns on microorganisms. Toll like receptors are a distinct class of pathogen recognition receptors.

TN_GOVT_XI_Micro_Biology_CH13.indd 232 16-02-2019 10.36.10 AM